

Type Casting

Def: converting a value from one data type to another.

How to use : `int a = 10;`
`double b = a;` } widening (small to big) - automatic

`double x = 9.7;`
`int y = (int) x;` } narrowing (big to small) - manual (explicit)

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when to use: when you need a value in a d/f type than it's currently stored in - e.g., dividing two integers but wanting a decimal result.

where to use: Calculations mixing types, converting user input, precision-sensitive operations.

why two kinds (widening vs narrowing): widening is safe (no data loss) so java does it automatically. Narrowing risks losing data (e.g., decimal truncation), so java forces you to explicitly write `(int)` to confirm you accept that risk. Real-life ex: like pouring a large bottle of water (double) into a small cup (int) - some water spills (data lost). pouring a small cup into a large bottle (int to double) - nothing is lost, fits easily.